UGANDA CHRISTIAN UNIVERSITY

A Centre of Excellence in the Heart of Africa

Faculty of Engineering, Design and Technology
Department of Computing & Technology

Agricultural Services Database

Project Documentation & Technical Report

Course Code: CSC2209 — Database Programming
Academic Year: Easter 2026
Programme: Bachelor of Computer Science (Year 2)
Group: ONE

Student Name	Registration Number
Mayinja Joel	S24B23/047
Mugoya Andrew	M24B23/013
Rockdit Minyiel Ayuak	M24B23/040
Mwesigwa Simon Peter	S24B23/102
Tumusiime Julius	S24B23/065
Ssendagire Abubaker	S24B23/012

Contents

1 Introduction **2**
1.1 Project Background . . . 2

2 Milestone 1: Requirements Specification **2**
2.1 Elicitation Methods . . . 2
2.2 Data Collection: Digital Questionnaire . . . 2

3 Milestone 2: Database Design (EERD) **4**
3.1 Design Methodology . . . 4
3.2 Enhanced Entity Relationship Diagram . . . 4
3.3 Business Rules . . . 4

4 Milestone 3: Database Development **5**
4.1 Development Environment . . . 5
4.2 Core Table Structures . . . 5
4.3 The Person Supertype Pattern . . . 5
4.4 Input and Distribution Management . . . 6
4.5 Summary of Applied Constraints . . . 6

5 Milestone 4: Security and Automation **7**
5.1 Database Triggers . . . 7
5.2 Stored Procedures and Transactions . . . 7
5.3 Reporting Views . . . 8
5.4 Role-Based Access Control (RBAC) . . . 8

6 Milestone 5: Testing and Validation **9**
6.1 Unit Testing Results . . . 9
6.2 Backup and Recovery Drill . . . 9
6.3 Lessons Learned . . . 9
6.4 Future Recommendations . . . 9

7 Conclusion **9**

1 Introduction

1.1 Project Background

The Ministry of Agriculture (MAAIF) lacks a centralized system to track coffee production and input distribution. The `agric_db` addresses this by providing a relational framework to monitor growers and manage agricultural resources efficiently.

2 Milestone 1: Requirements Specification

2.1 Elicitation Methods

We gathered requirements through stakeholder interviews, document analysis of Ministry templates, and field observations.

2.2 Data Collection: Digital Questionnaire

To validate farmer needs, a digital questionnaire was distributed. This tool allowed us to capture data on farmer education levels and years of experience to better tailor extension services.

MINISTRY OF AGRICULTURE

National Coffee Farmer Database – Data Collection Questionnaire

To be filled by farmer or enumerator – all information is confidential

Instructions: Please answer all questions as accurately as possible. For sections with multiple entries (e.g., several farms or production seasons), use the tables provided. Use an additional sheet if needed. Mark where applicable.

A. Farmer Registration (One record per farmer)

Full name Namulondo sarah

Gender Male Female Other

Date of birth (dd/mm/yyyy) 14/11/72

National ID / Cf720081029v2g

Voter ID

Phone number 07o4462701

District Mukono M

Sub-county / Nama

Parish

Village / Cell Katoogo Waluga

Nearest town Katoogo

B. Farm Details (At least one farm – add all farms you own)

Use one row per farm. *If you own more than 3 farms, continue on a separate sheet.*

Farm #	Farm name / identifier	Location (village, landmark, or GPS)	Area (acres / hectares)	Approx. # of coffee trees
1	Saramugo	Waluga 9	2000	
2				
3				

C. Coffee Production per Season and Year

Record the total harvest (fresh cherry or parchment, in kilograms) for each completed season. *If more than 4 seasons, use extra sheet.*

Figure 1: Questionnaire used for farmer data elicitation

3 Milestone 2: Database Design (EERD)

3.1 Design Methodology

The design uses Enhanced Entity Relationship (EER) modeling, specifically implementing Generalization and Specialization. The `Person` entity serves as a supertype with disjoint subtypes for Farmers, Extension Workers, and Ministry Officials.

3.2 Enhanced Entity Relationship Diagram

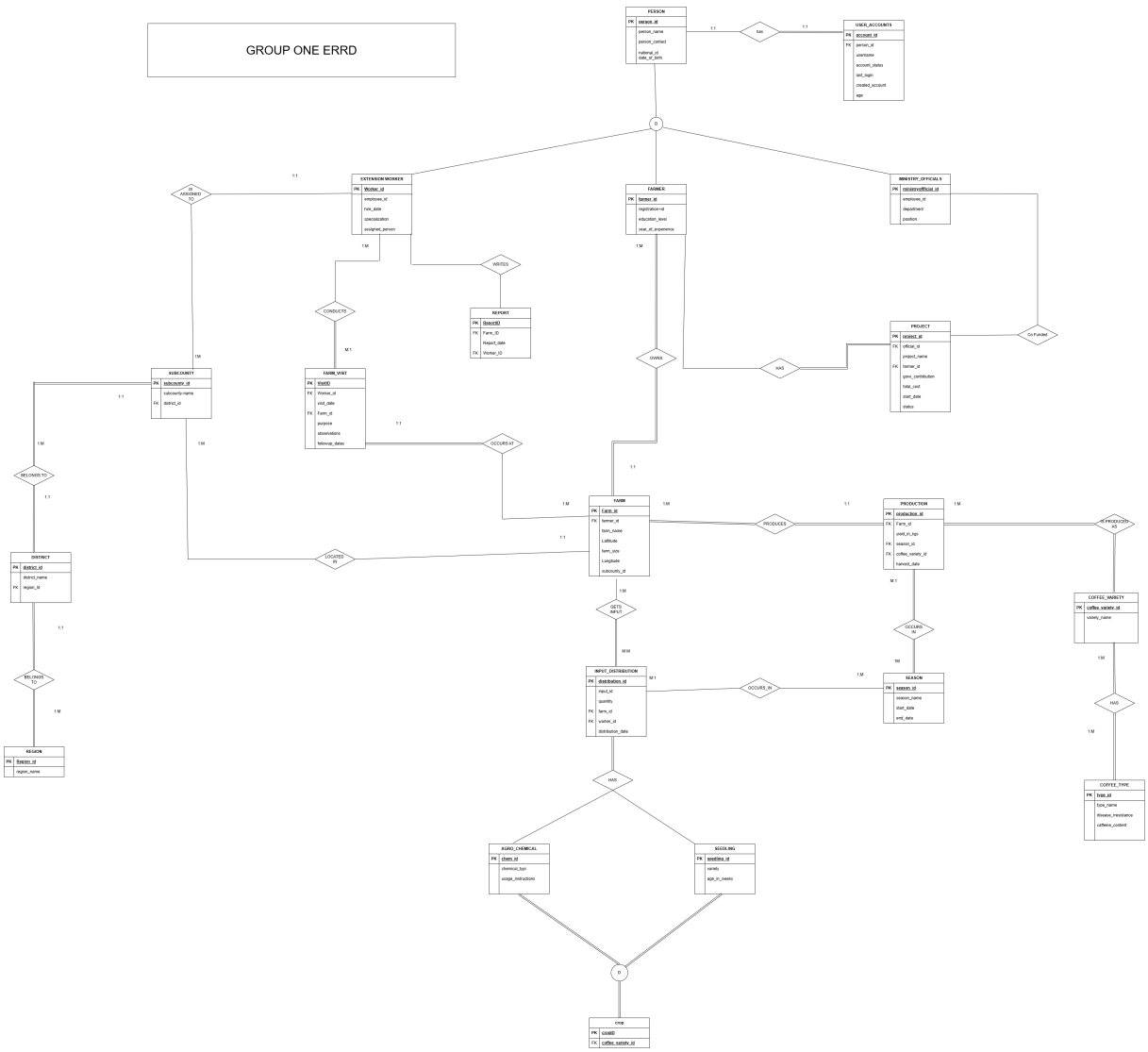

Figure 2: Enhanced Entity Relationship Diagram (EERD) of the Agricultural Database

3.3 Business Rules

- Each **Person** has exactly one User Account and may be a Farmer, Extension Worker, or Ministry Official.

- A **Farmer** can own multiple Farms; each Farm belongs to one Farmer.
- A **Farm** produces multiple Production records; each Production belongs to one Farm and occurs in one Season.
- A **Farm** is located in one Subcounty; Subcounties belong to Districts, and Districts belong to Regions.
- An **Extension Worker** is assigned to one Subcounty and conducts Farm Visits.
- Each **Farm Visit** is linked to one Farm and conducted by one Extension Worker.
- **Reports** are written by Extension Workers and linked to farm activities.
- **Input Distributions** supply Farms with inputs and are handled by Extension Workers.
- Inputs are categorized as **Seedlings** or **Agrochemicals**.
- A **Production** record references one Coffee Variety and occurs in a Season.
- A **Coffee Variety** belongs to one Coffee Type.
- A **Project** involves Farmers and may be co-funded.

4 Milestone 3: Database Development

4.1 Development Environment

The system was implemented using MySQL 8.0 within a Visual Studio Code IDE environment, utilizing XAMPP for local server hosting.

4.2 Core Table Structures

The following SQL defines the hierarchical geographic structure and the primary identity tables.

```sql
CREATE TABLE Region (
    region_id INT AUTO_INCREMENT PRIMARY KEY,
    region_name VARCHAR(100) NOT NULL UNIQUE
);

CREATE TABLE District (
    district_id INT AUTO_INCREMENT PRIMARY KEY,
    district_name VARCHAR(100) NOT NULL UNIQUE,
    region_id INT NOT NULL,
    FOREIGN KEY (region_id) REFERENCES Region(region_id) ON
       DELETE CASCADE
);
```

Listing 1: Geographic Hierarchy Implementation

4.3 The Person Supertype Pattern

To maintain referential integrity, we used a PK-FK pattern where subtypes inherit identity from the `Person` table.

```sql
CREATE TABLE Person (
    person_id INT AUTO_INCREMENT PRIMARY KEY,
    person_name VARCHAR(150) NOT NULL,
    national_id VARCHAR(20) UNIQUE
);

CREATE TABLE Farmer (
    farmer_id INT PRIMARY KEY,
    registration_id VARCHAR(50) UNIQUE NOT NULL,
    year_of_experience INT CHECK (year_of_experience >= 0),
    FOREIGN KEY (farmer_id) REFERENCES Person(person_id) ON
       DELETE CASCADE
);
```

Listing 2: Person Supertype and Farmer Subtype

4.4 Input and Distribution Management

The database tracks both seedlings and agro-chemicals via an `Input` supertype.

```sql
CREATE TABLE InputDistribution (
    distribution_id INT AUTO_INCREMENT PRIMARY KEY,
    input_id INT NOT NULL,
    farm_id INT NOT NULL,
    worker_id INT NOT NULL,
    quantity DECIMAL(10,2) NOT NULL CHECK (quantity > 0),
    distribution_date DATE NOT NULL,
    FOREIGN KEY (input_id) REFERENCES Input(input_id),
    FOREIGN KEY (farm_id) REFERENCES Farm(farm_id)
);
```

Listing 3: Input Distribution and Junction Tables

4.5 Summary of Applied Constraints

A variety of constraints were applied to enforce business logic:

- **UNIQUE**: Applied to National IDs and Employee IDs to prevent duplicates.
- **CHECK**: Used to ensure farm sizes and experience years are non-negative.
- **ENUM**: Restricts account statuses and input types to predefined sets.

5 Milestone 4: Security and Automation

5.1 Database Triggers

Triggers handle silent automation. For instance, `trg_Set_Record_Date` ensures every production entry has a timestamp if none is provided.

```
CREATE TRIGGER trg_Prevent_Negative_Production
BEFORE INSERT ON Production
FOR EACH ROW
BEGIN
    IF NEW.yield_in_kgs < 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Error: Production quantity cannot be
           negative.';
    END IF;
END;
```

Listing 4: Data Validation Trigger

5.2 Stored Procedures and Transactions

We utilize stored procedures to wrap multi-table logic into atomic transactions, ensuring data consistency.

```
CREATE PROCEDURE sp_AddFarmer(...)
BEGIN
    START TRANSACTION;
        INSERT INTO Person (...) VALUES (...);
        SET new_person_id = LAST_INSERT_ID();
        INSERT INTO Farmer (farmer_id, ...) VALUES (new_person_id
           , ...);
    COMMIT;
END;
```

Listing 5: Atomic Farmer Registration

5.3 Reporting Views

Views are utilized to hide sensitive data like National IDs from general ministry staff while providing useful summaries.

- **vw_ActiveFarmers**: Shows only farmers who harvested in the last 12 months.
- **vw_LowProductionFarmers**: Flags farmers producing under 500kg for support.

5.4 Role-Based Access Control (RBAC)

Access is limited based on the "Principle of Least Privilege".

User Role	Privileges
System Admin	Full schema control and ALL PRIVILEGES.
Ministry Official	SELECT on views; INSERT/UPDATE on Projects.
Extension Worker	Access only to FarmVisit, Report, and InputDistribution.
Farmer App	EXECUTE privilege only on specific procedures.

Table 1: Summary of Database RBAC Policy

6 Milestone 5: Testing and Validation

6.1 Unit Testing Results

The system was subjected to 8 distinct test cases to verify constraints and automation logic].

ID	Description	Result
TC-01	Add farmer with valid data (Atomic Check)	PASS
TC-03	Insert duplicate National ID	PASS
TC-04	Insert negative production yield	PASS
TC-07	Extension worker attempts to DELETE farmer	PASS

6.2 Backup and Recovery Drill

To ensure resilience, we established a daily automated backup schedule. A monthly restoration drill is conducted to a test server to verify the integrity of the `.sql` dump files.

Recovery Time Objective (RTO): Under 2 hours.
Command: `mysqldump -u root -p agric_db > full_backup.sql`.

6.3 Lessons Learned

- **Design First**: Rigorous EER modeling before SQL implementation reduces structural rework.
- **Transaction Safety**: Wrapping multi-table operations in transactions prevents "orphan" records in the supertype tables.
- **Hidden Triggers**: Triggers are powerful for audit logging but require careful documentation as they fire silently.

6.4 Future Recommendations

- **Geospatial Mapping**: Integrate farm GPS coordinates with a Mapping API for better visual oversight.
- **Audit Trails**: Implement a dedicated audit log to track "who changed what and when" for full accountability.

7 Conclusion

The `agric_db` project successfully delivers a functional system with 18 tables, 3 triggers, and 3 stored procedures. By enforcing data integrity and role-based security, the Ministry of Agriculture is now equipped to digitalize the coffee sector data management efficiently.